Cloud Computing (cloud)

Storage

Example Exam-Questions

1. Multiple Choice:
   - Fix defined scenario e.g. “map the following technologies to the layers IaaS/PaaS/SaaS-ILayers”.
     + Points for right answers / - Points for wrong answers / not less than 0 points in this question possible
2. Comprehesive/Knowledge Questions:
   - What is the purpose of the kube-apiserver/etcd/kubelet? What happens if it is not available any more?
     Might cover lecture, tutorial and chapters from reading list
3. Sourcecode:
   - Given the following log outputs: What can be the failure of the system (e.g. not synced nodes within Ceph)
   - Given following Dockerfile, annotate the different layers and describe what they do.
     Might focus more on tutorials.
4. Graphical Questions:
   - Annotate the following (empty) components describing a control plane in K8s.

MSP, Date and Infos

19.12.25: 8:45-9:45h

- 6.0D13

Materials:

- MSP-Summary.docx from Teams
  - At most 1 page per session / week
  - Will be provided by us in a printed form
  - Feel free to collaborate on the content, organize as you want (using the provided teams channel, etc.)
  - Deadline for writing to the document: 7.7.25, 11:00am
- No further materials are allowed

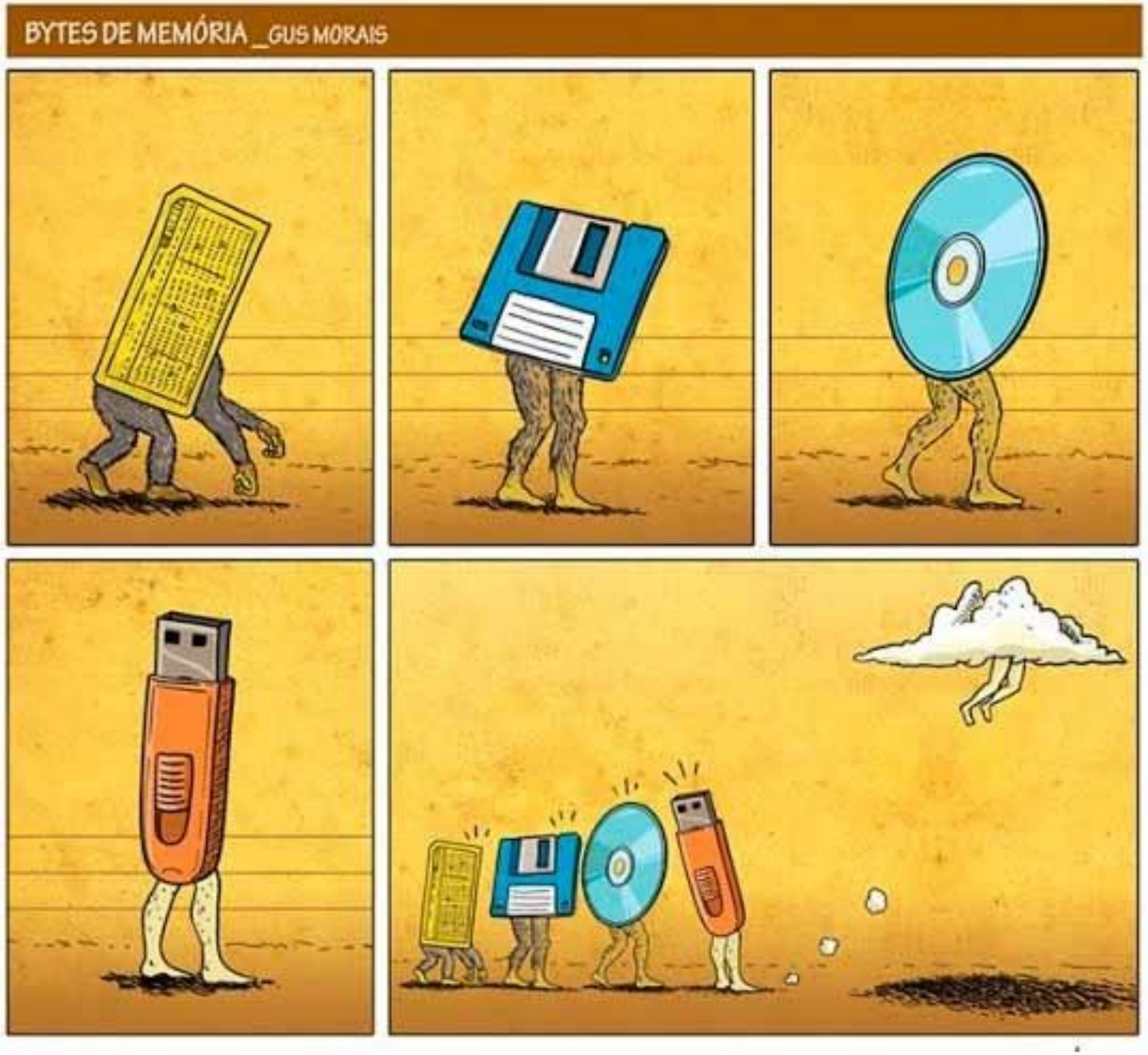
Platform:

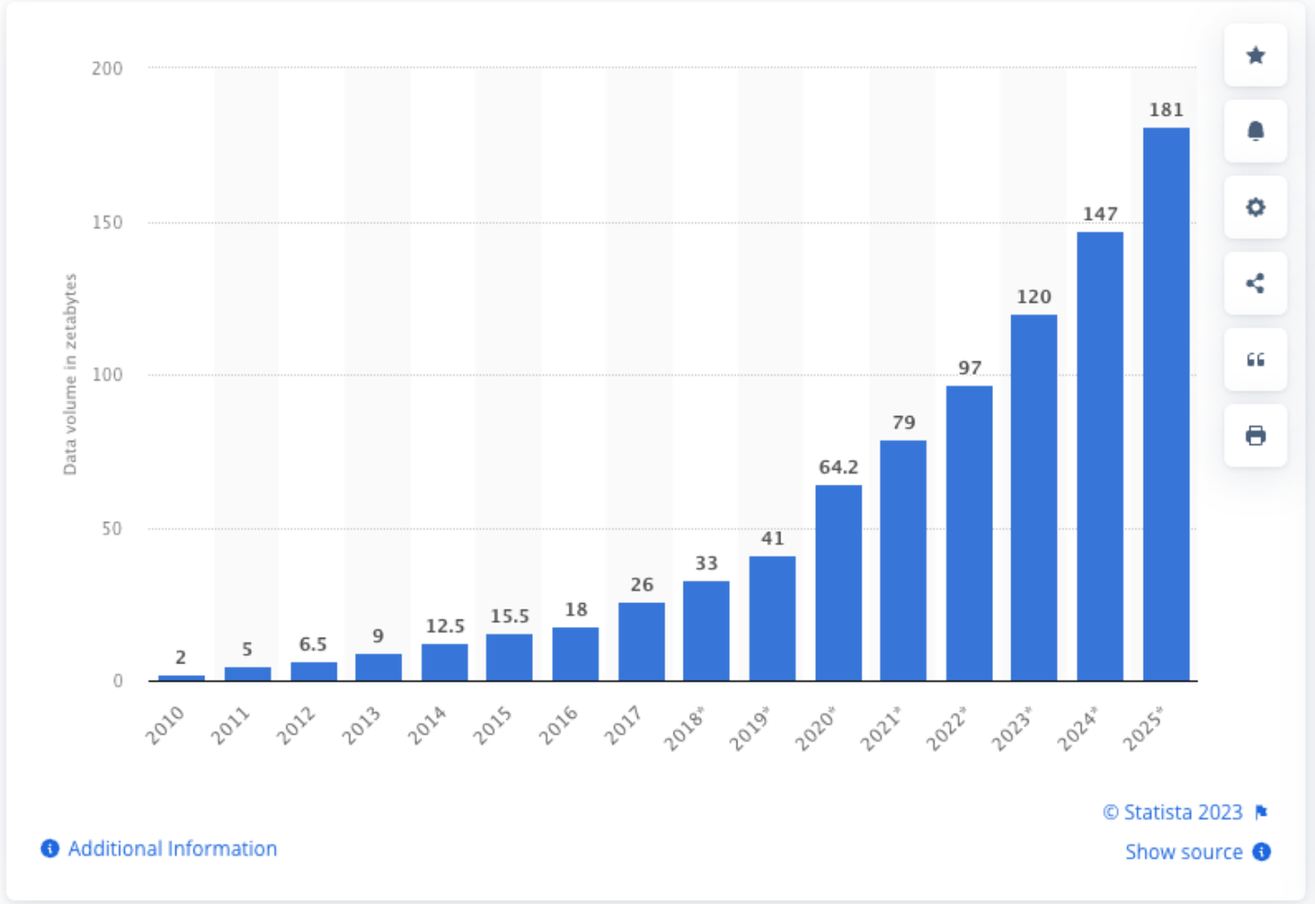
- Moodle with LearnStick / SafeExam Browser
  - No Booting, just SafeExam Browser
- Test Exam → 12.12.25
  - You are responsible to make sure your infrastructure works with the setup, this is your chance to check it.
  - Check is not able to be performed remotely.

Custom Resources

- https://learning.oreilly.com/playlists/43827098-7a87-435e-823e-736586b5694c
  10-storage
- Several Blogposts
  - CERN: https://indico.cern.ch/event/649159/contributions/2761965/attachments/1544385/2423339/hroussea-storage-at-CERN.pdf
- Ceph-Documentation: https://docs.ceph.com/
- Kubernetes-Documentation: https://kubernetes.io/docs/concepts/storage/persistent-volumes/#volume-mode

Evolution of Storage





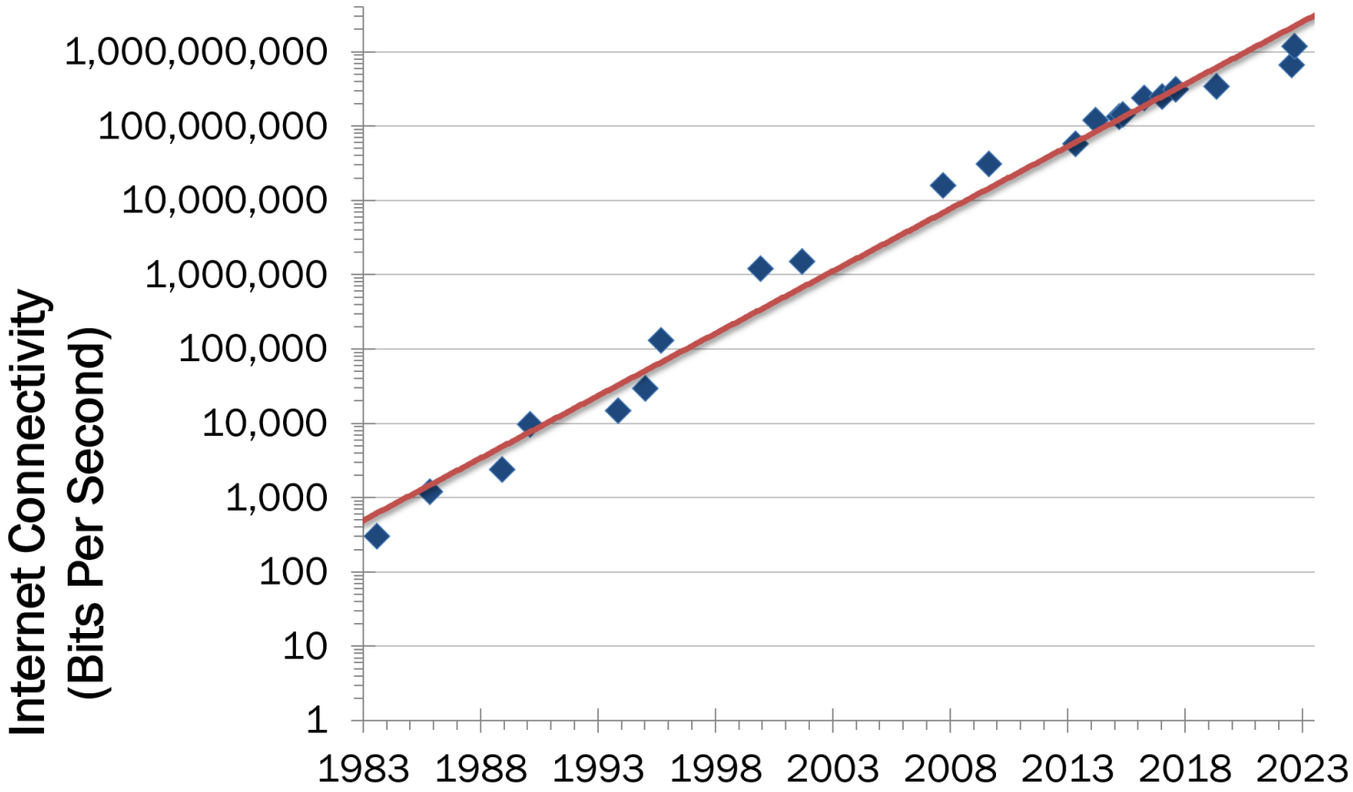
Why do we need Storage?

Just reflect the immense amount:

- 282 pages → 1MB
- 536 books → 1GB
- 549755 book → 1TB
- 192 Library of Congress → 1PB
- 1000000 PB → 1 ZB

Nielsen's Gesetz: "High-end users connection speed increases by 50% per year."

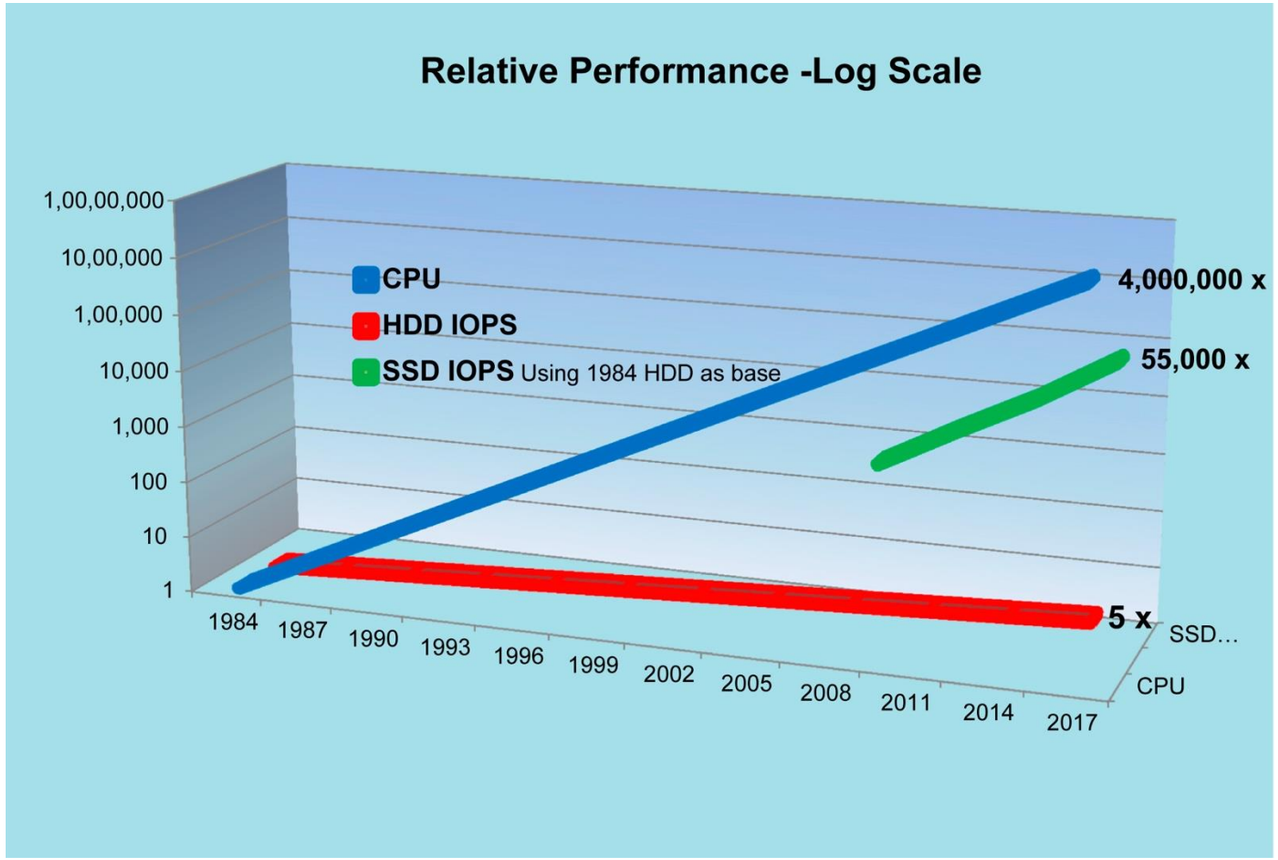
Why Storage matters? → Bandwith increases



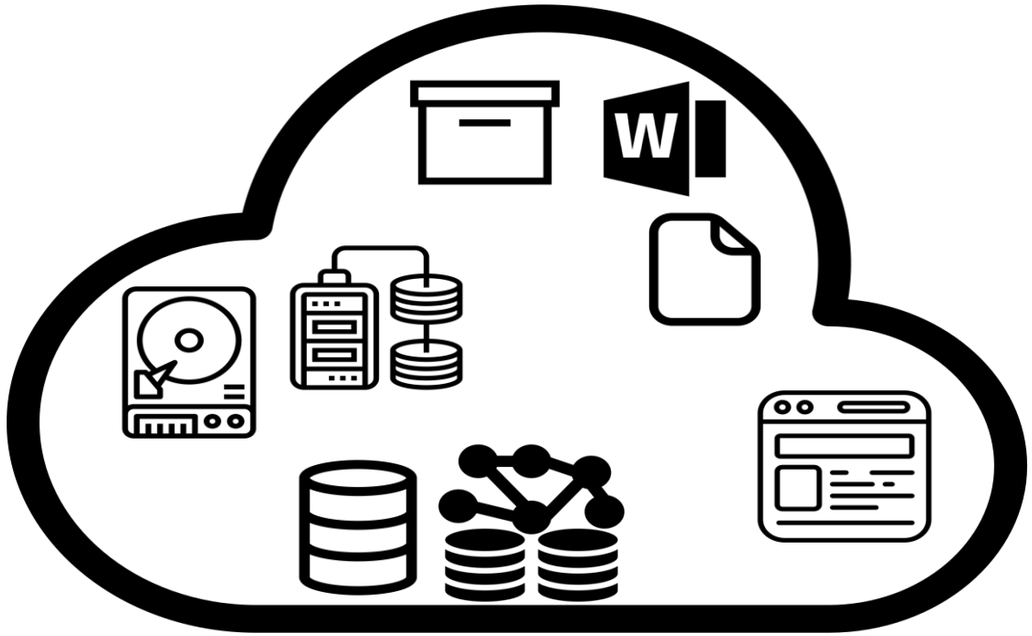
No need to be directly near to computations:

- Bandwith increases massively
- Architectures tackle latency
- Software organizes access

Why Storage matters?
→ Secondary Storage keeps up with Moores' Laws



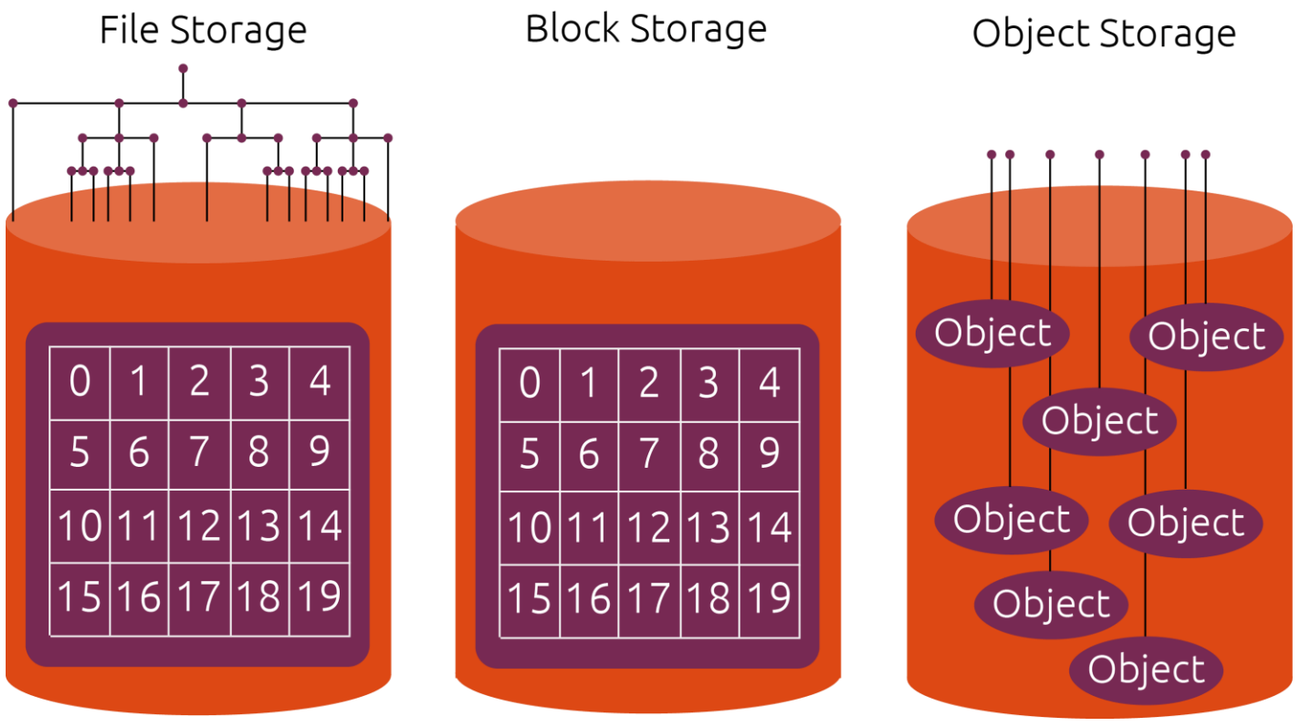
- IOPS of SSD increase the same scale as CPU-cycles
- At the same time, SSD prices become lower than HDD prices for enterprise hardware
- IOPS more important than plain bandwith



Cloud is Service First

- Disk-based Storage → Blocks
- File-based Storage → Folders / Buckets / Filesystems
- Structured Storage → (No)SQL Datastores
- Displaying Storage → Buckets, SaaS-Services of any kind

What is Storage?



Block Storage:

- Addressing storage areas on disk (”blocks”)

File Storage:

- Addressing files in a filesystem on disks

Object Storage:

- Addressing “objects”, which are technically based on files

What is a Blockstorage?

Blocks are just a fixed sized area of data on the disk.

Unique Identifies of.a block are:

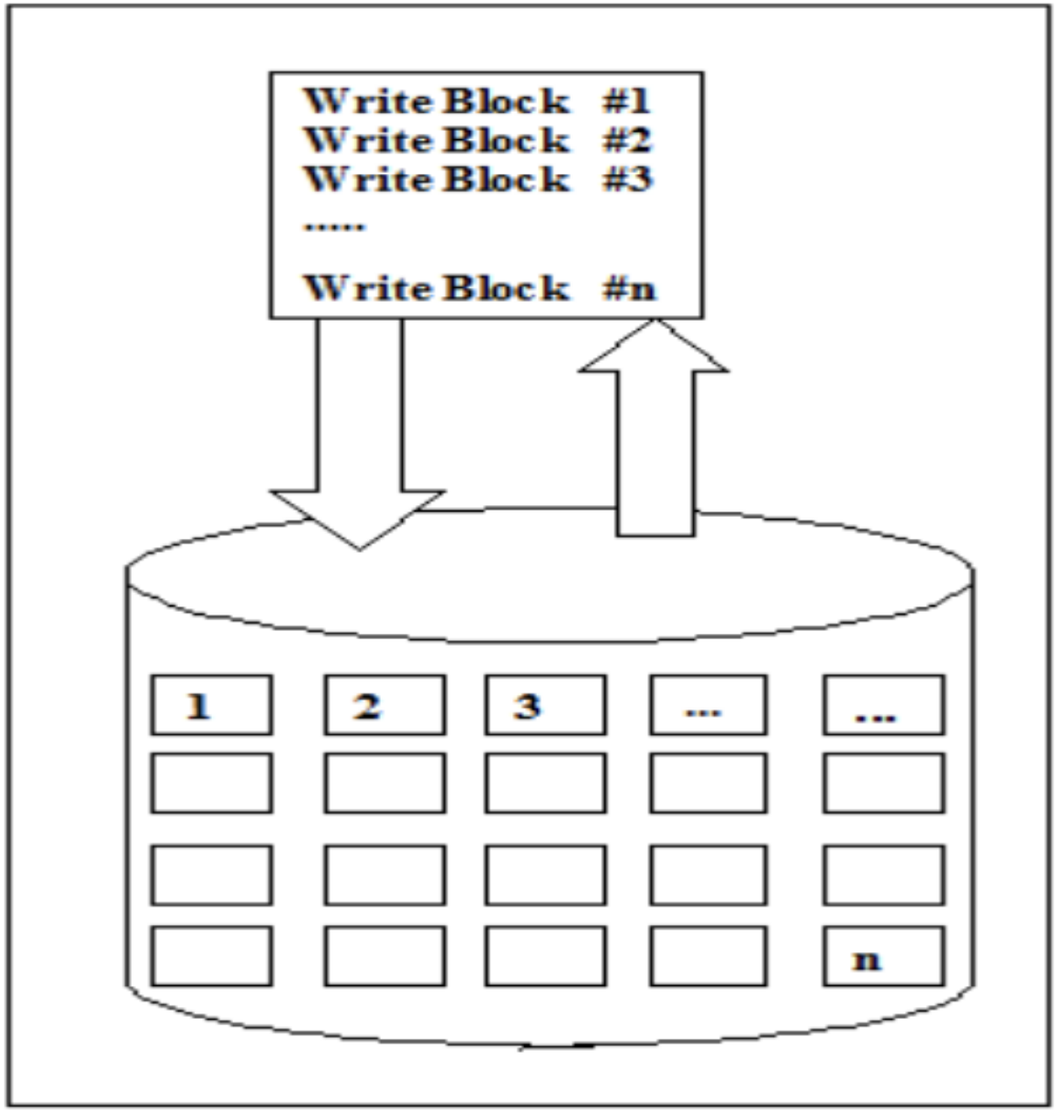
- Offset on the disk
- Length of the block (usually all blocks have the same length

→ Operated by Operating System

→ No Index / Metadata: Everything is a block

→ Blocks may have pointers to other blocks (remember Inodes?)

→ Accessing Protocols are iSCSI, FibreChannel, SATA, NVMe…



Accessing Block Storage, iSCSI

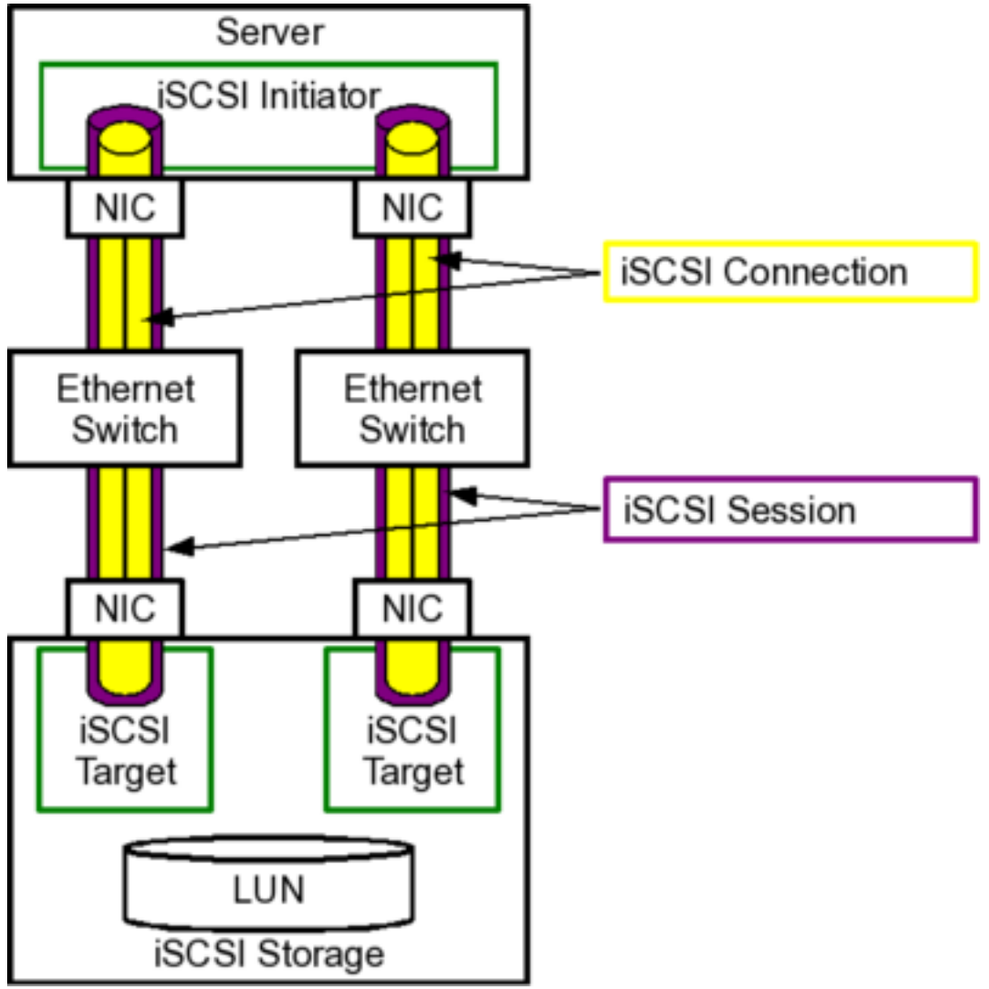
Access to blocks take place via Server/Client Setup and plain TCP/IP (Port 3260):

- Server → iSCSI target
- Client → iSCSI initiator

Target and Initiator can be in Software or Hardware:
→ In Software, they are often part of the OS

To access, a session and a connection must be used:

- Session: permament setup between initiator and target, used for lookup and init
- Connection: Single access for reading and writing blocks (offset / length) within an existing session



Benefits / Drawbacks of Blockstorage

Benefits:

- Flexibel, eigenes FS
- Einfache Verschlüsselung (symentrisch)
- Vergleichsweise schnell
  (wenig applikatorischen Overhead)

Drawbacks:

- Kein Lesen der gesamten Blockgrösse
- Nicht direkt für (die meisten) Applikation nutzbar
- Keine fein granulare Authorisierung auf einzelne Blöcke

What is a Filestorage?

Files are elements of a filesystem, bringing structure in the chaos of blocks.

- Overlaying structure of blocks necessary (e.g. INodes in Linux Systems)

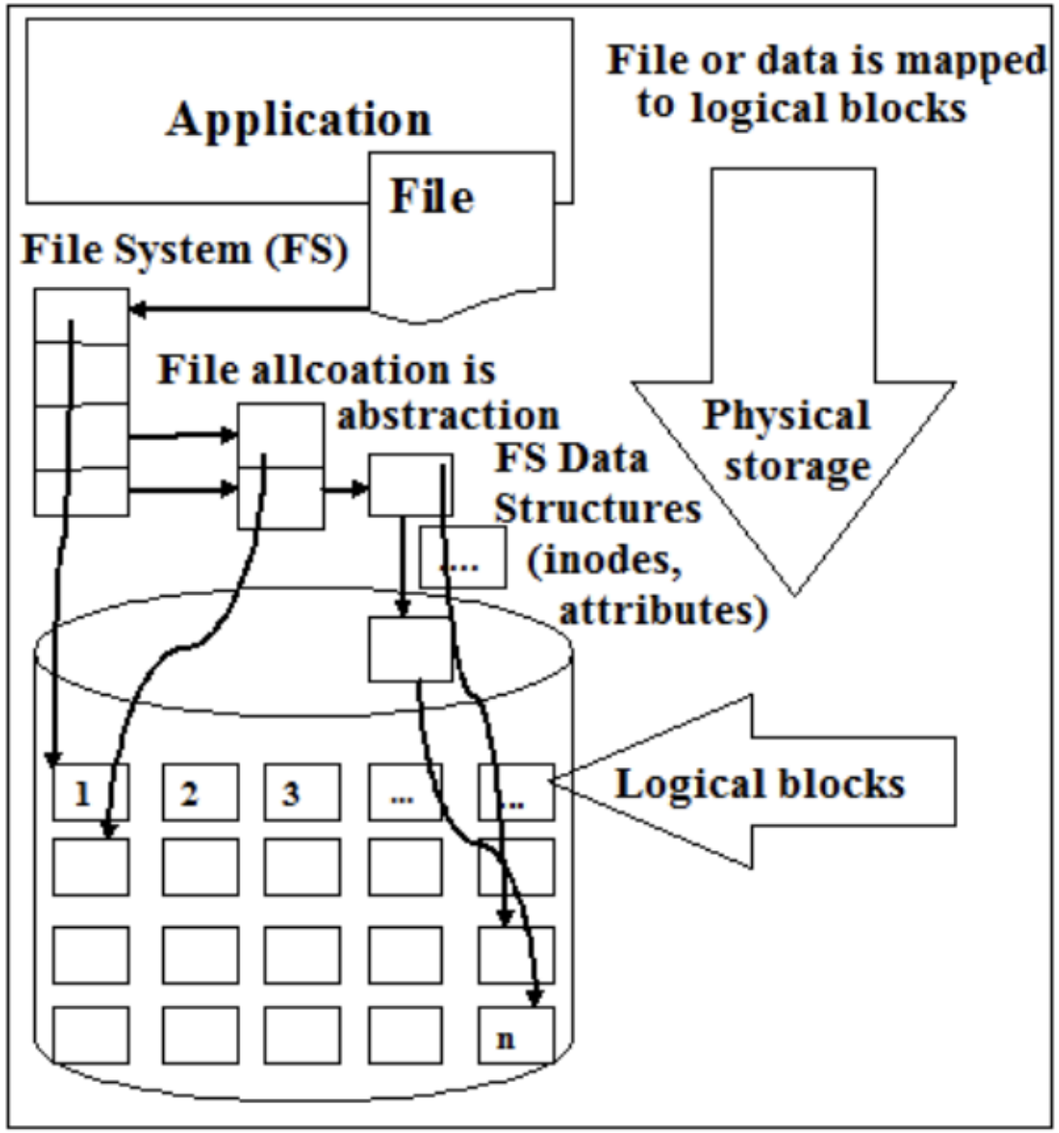
→ Variable length of elements

→ Depending on OS

→ Internally hierarchically structured (mapping partially the structure of block pointers)

→ Common Filesystems are ext[2-4[, btrfs, ntfs, apfs

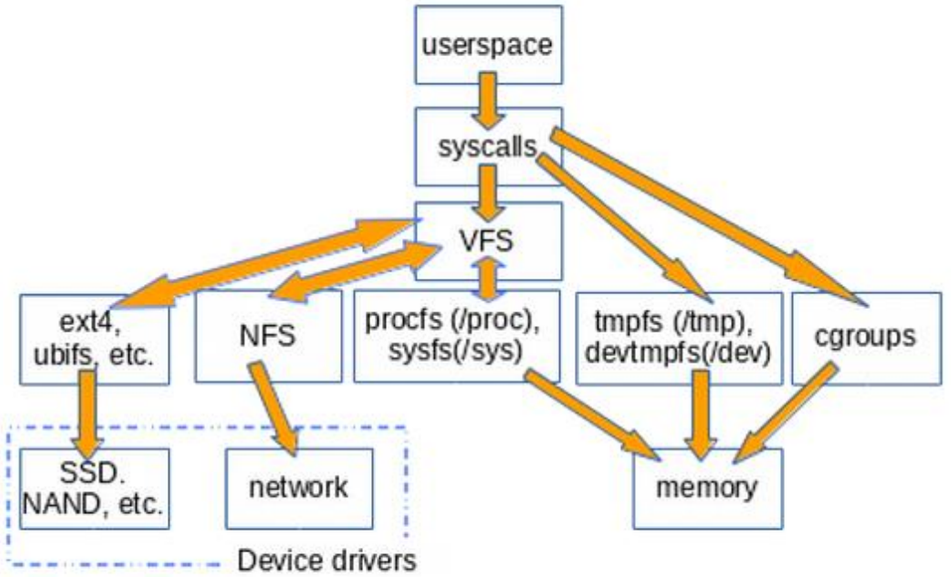
→ Accessing Protocols are SMB, SSH, NFS, HTTP (Webdav)



Accessing Files / VFS

Filesystems bring order in the list of blocks:
(In fact, they are organizing entire operating systems e.g. with VFS):

- Flexible sized elements (thanks to the Inode Structure)
- Owner / Access Management of Elements
- Links
- Further OS-Management:
  - Processorganization
  - Ressourcemanagement (remember cgroups?)
  - Devicemanagement



Benefits / Drawbacks of Filestorage

Benefits:

- Direkter Zugriff auf Daten von Applikation nötig mit am kompatibelsten für Anwendungen
- Dahinterliegende Logik ist nicht in der Applikation
- Permissions / Rechtemgmt / Authorisierung möglich
- Integritätsschutz / Journaling / Snapshots

Drawbacks:

- Jedes Dateisystem hat Flaws
- Overhead im Lesen/Schreiben
  - Einhergehend mit Funktionalität
  - Einhergehend mit der Blockstruktur
- OS-abhängig

What is a Object Store?

– Objects are larger binary data referenced by a unique key:

- Flat structure, no hierarchy

→ Variable length of values

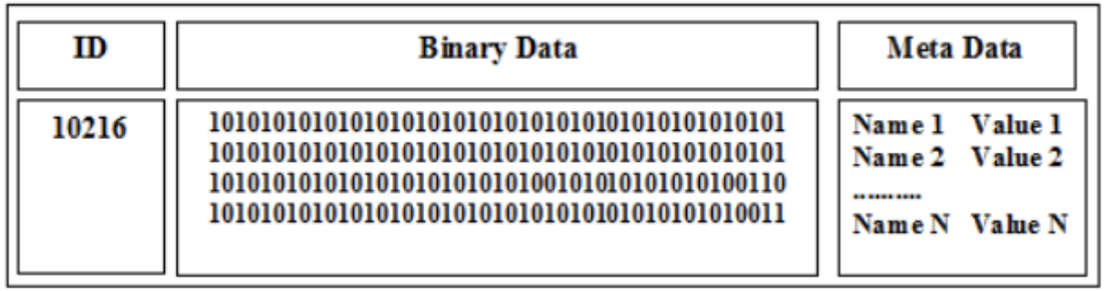
→ Independing on OS, used by applications often

→ Can be enriched by metadata

→ Can be implemented by a filesystem with flat hierarchy as backend

→ Recommended Pattern: Write once, read often

→ Really becoming popular due to the cloud move since application can easy access data directly most often based on HTTP



Accessing Object Stores / REST

example URL format	GET	POST	DELETE
<object_type>	returns object list (query, filters) 200	creates new object OR makes bulk update 201 / 200	bulk delete (query, filters) 200
<object_type>/<id>/	returns single object, object-specific filters 200	updates single object 200	deletes single object 200
<object_type>/<id>/acl/	returns object permissions 200	updates object permissions 200	Not supported 405

- Request: GET URL/<object>/[<subobject_id>]/[?<params>]
- Request: DELETE URL/<object>/[<subobject_id>]
- Request: POST URL/<object>/[<subobject_id>]

Benefits / Drawbacks of Objectstorage

Benefits:

- “einfach” damit zu arbeiten:
  Standardmittel des Webs werden verwenden
- Einfacheres Exponieren der Dateien,
  transparentere Authorisierungsmechanismen
- Skalierung

Drawbacks:

- Applikation muss es unterstützen
  (analog zu Blockdevice)
- Grosser Overhead bei Lese-/Schreibzugriffe
- Limitierter Anwendungsbereich

SAN, DAS, NAS

DAS:

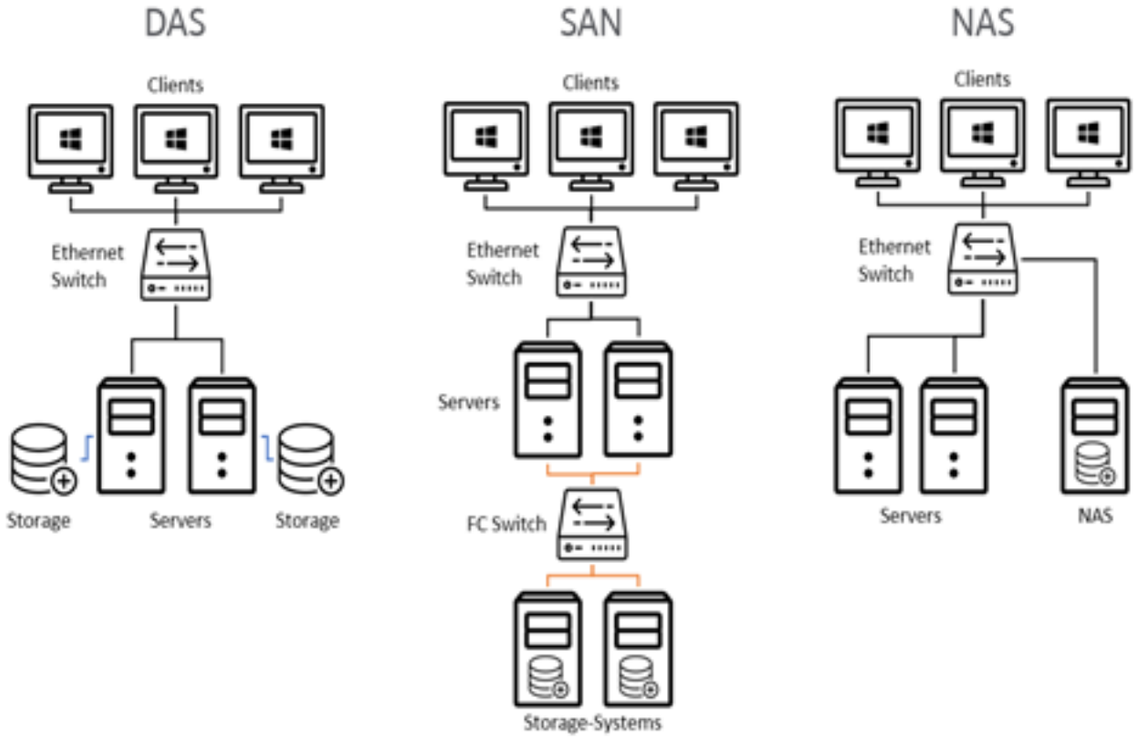
- Disk directly attached to server via SCSI / IDE/ SATA
- Cheap, easy, fast setup
- If server goes down, disk go down

SAN:

- Disk attached via FibreChannel to one/multiple servers
- Complex, second stack, more expensive
- Failover / Scaling possible

NAS:

- Disk attached directly as device to network
- Easier technical setup, cheaper, redundant
- Blockaccess needs workaround e.g. iSCSI

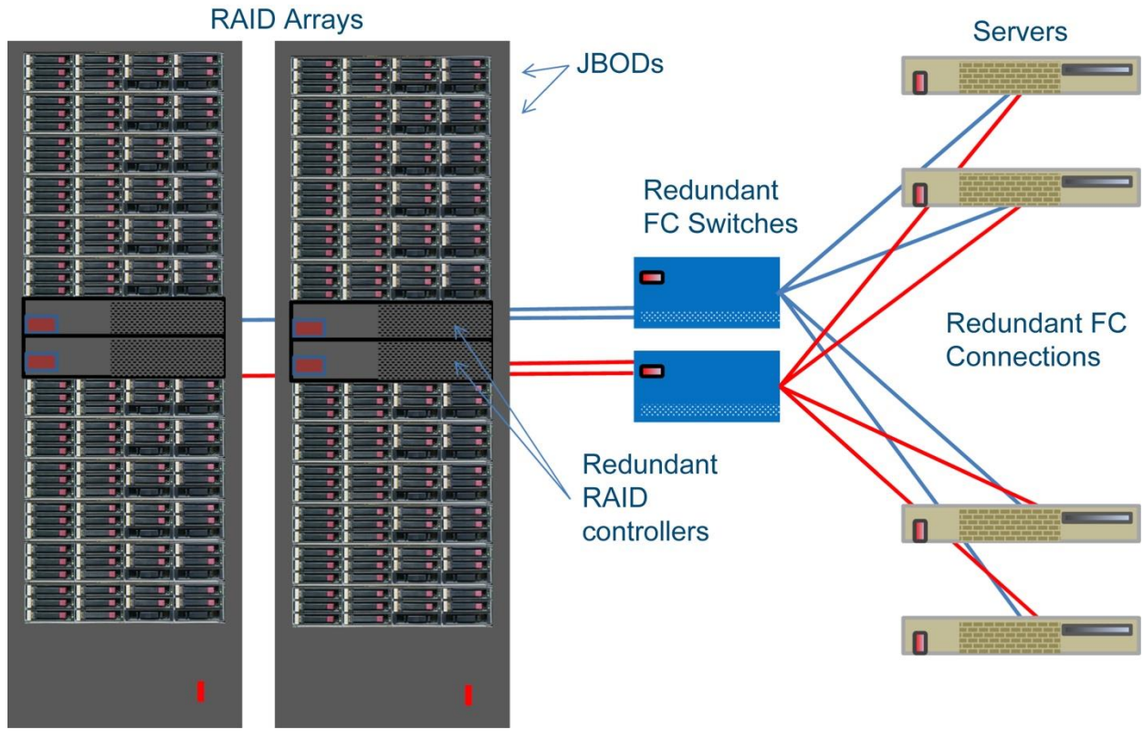


Example: Classical SAN

JBOD: Just a bunch of disks

→ Arranged in RAIDs
(Redundant Array of Inexpensive Disks)

→ Accessible via Fibre Channel Switch
(can be a normal switch when working with iSCSI:
→ RAID-Arrays represent iSCSI targets in that case)

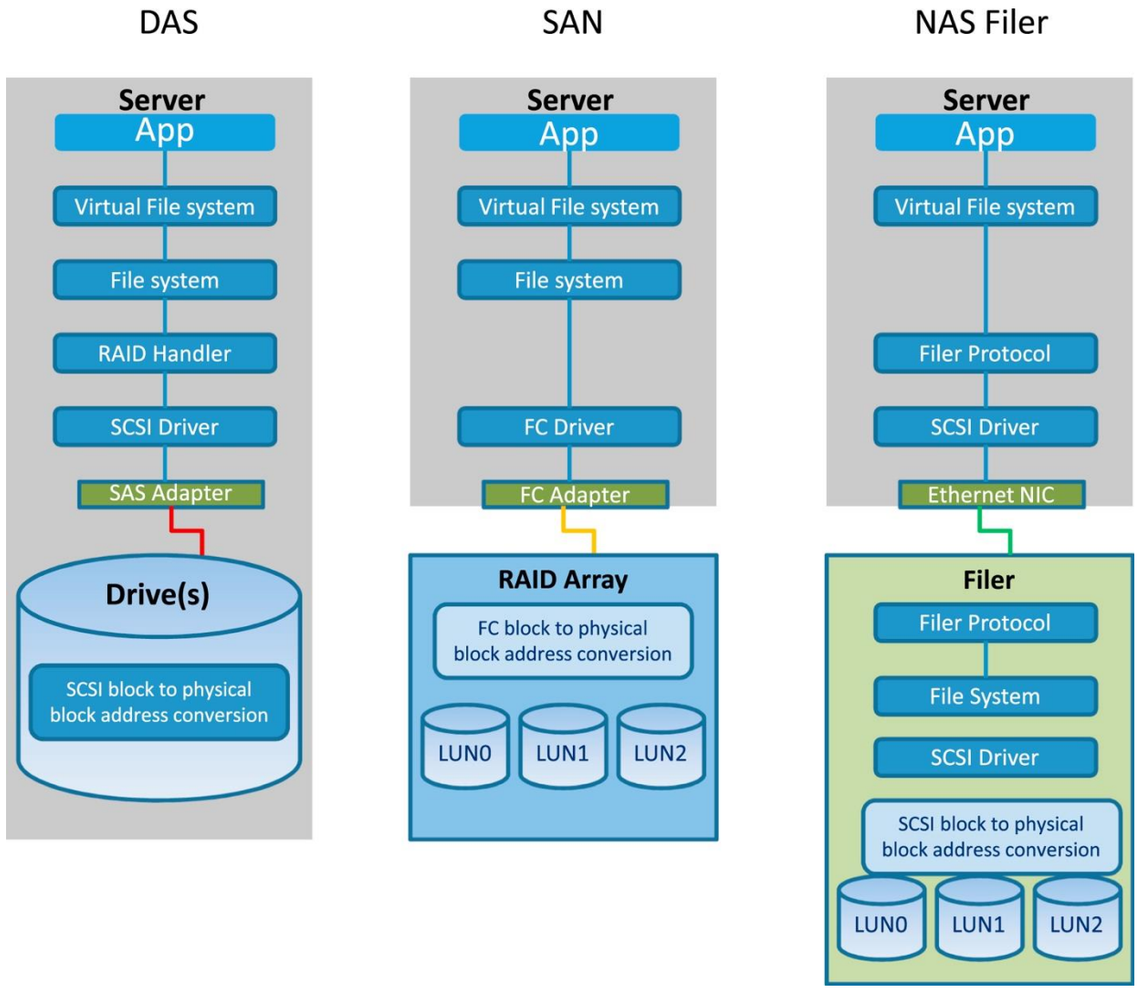


Problem: hardware restriction
(capacity / availability / etc.)

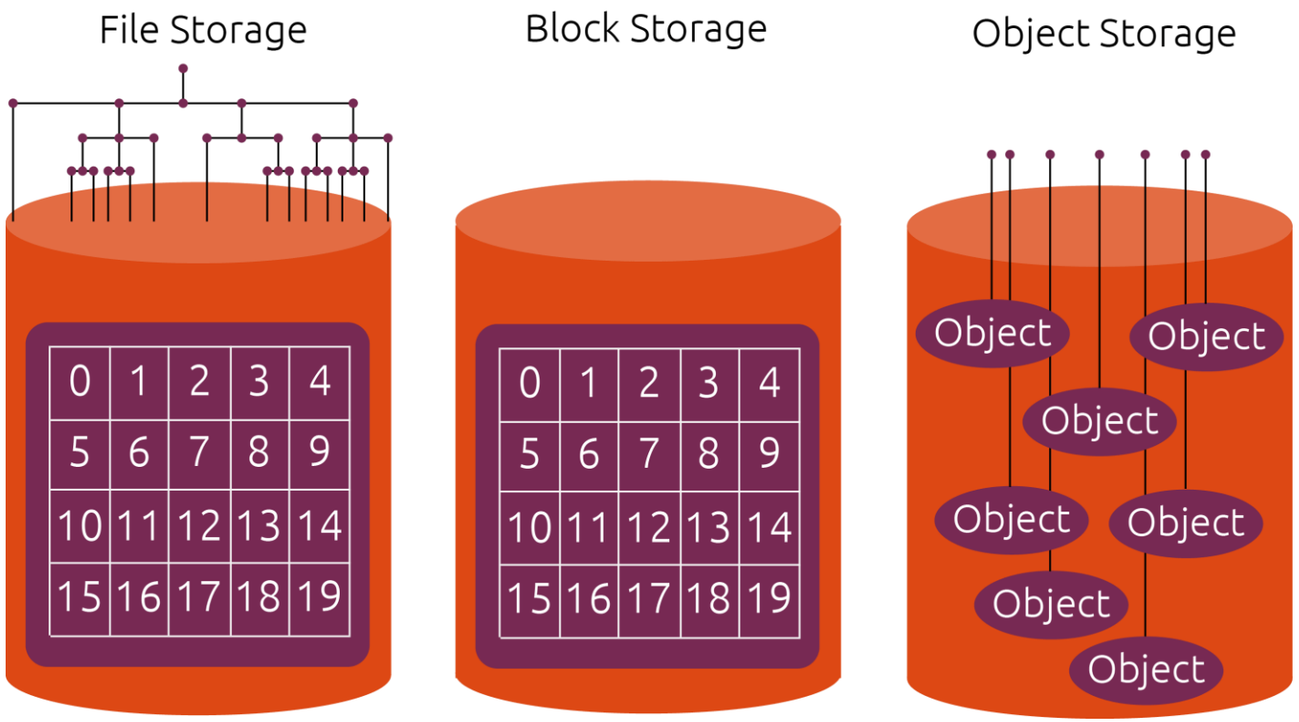
Mapping to storage kind

Level of abstraction so far is Bockstorage

- DAS/SAN: Metadata of file system is on disk themselves
  - (-) iterating over hierarchy needs to traverse the entire TCP/IP-Stack (remember OS-Layers)
- NAS: Metadata is on nodes themselves
  - Iterating through structure is faster
    - (-) Filer / NAS-nodes have own OS/Kernel with all complexity



Reminder: What is Storage?



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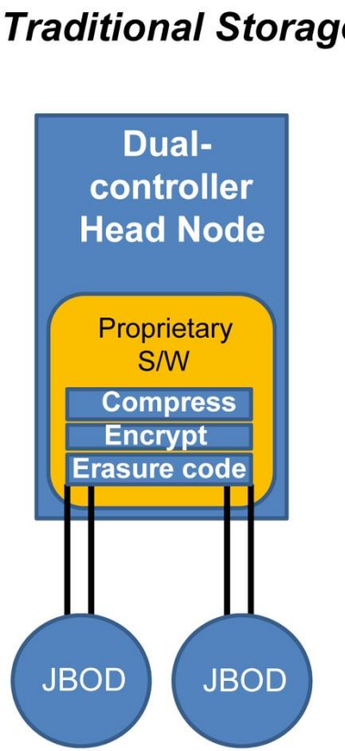
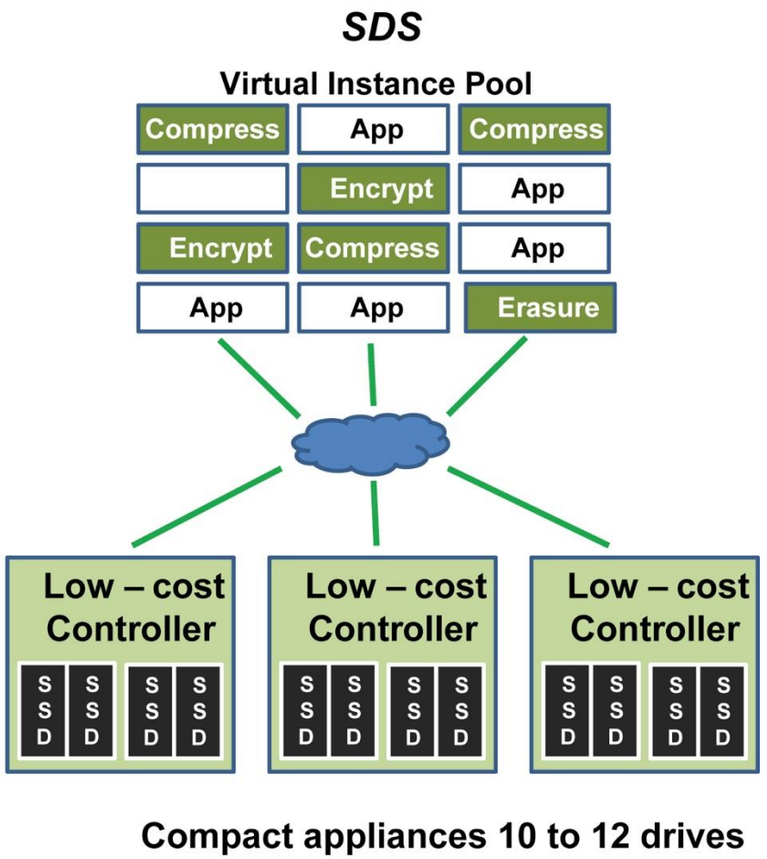
Software defined Storage to the rescue

SDS ”virtualizes” the storage

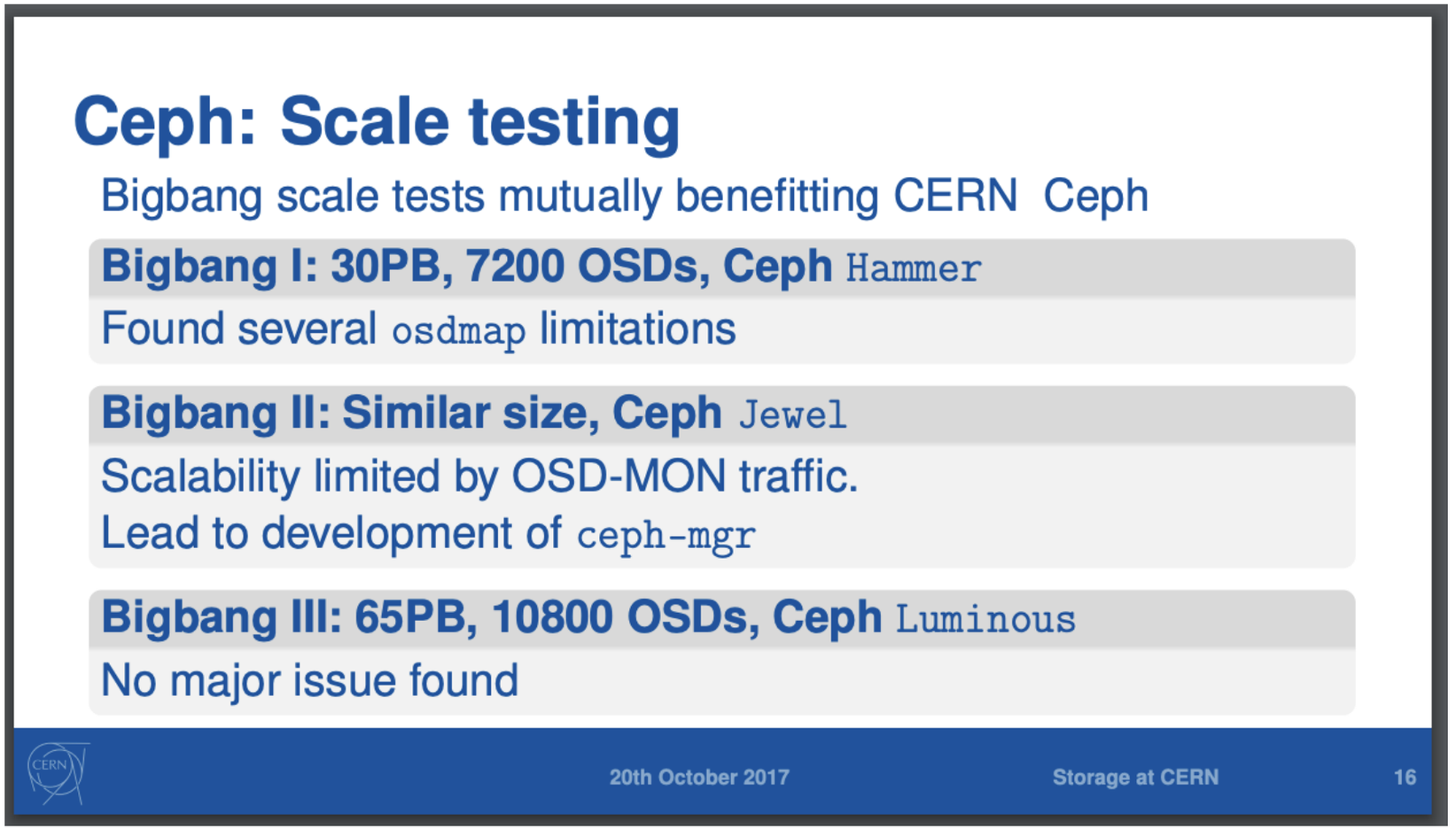
- Storage accesses are stacked before ”storage” (compress & erasure / compress & encrypt)
- Access to concrete storage is abstracted
  - Several machines offer software access to storage
  - Storage layer is entirely independent from data to read/write
  - HA/extension is easier

What is the problem?





SDS-Example in practice: Ceph



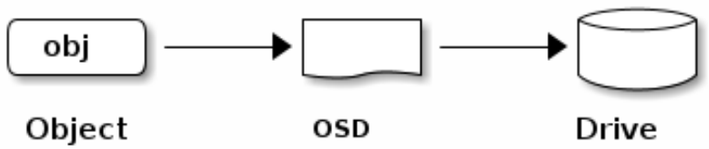
https://indico.cern.ch/event/649159/contributions/2761965/attachments/1544385/2423339/hroussea-storage-at-CERN.pdf

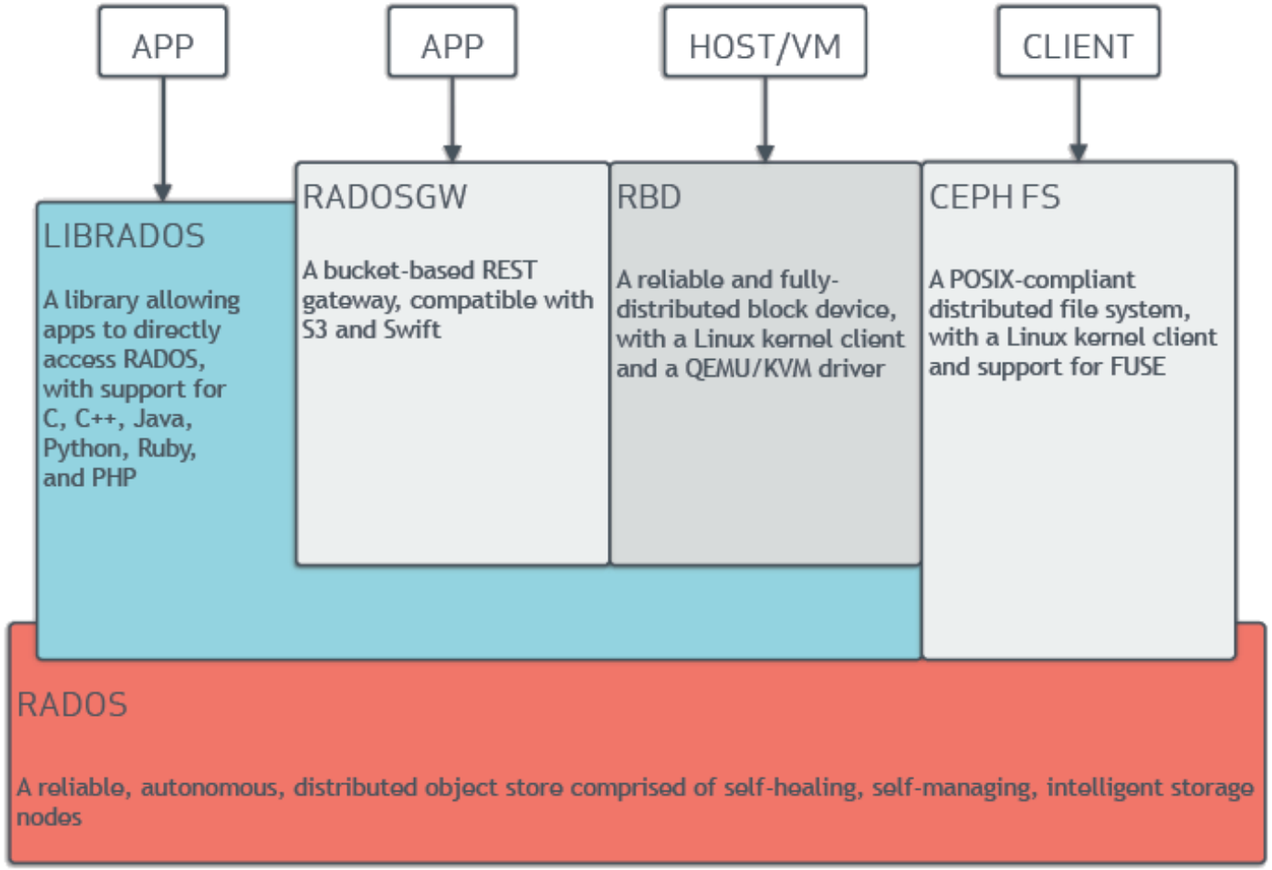
Concrete Example of SDS, CEPH

Offering interfaces for:

- Block: RBD = iSCSI target
- File: CephFS as Filesystem
- Object: RadosGW as Bucketstorage

– All data is transferred to a OSD (Open Stack Demon) mapping drives on blocklevel





OSDs

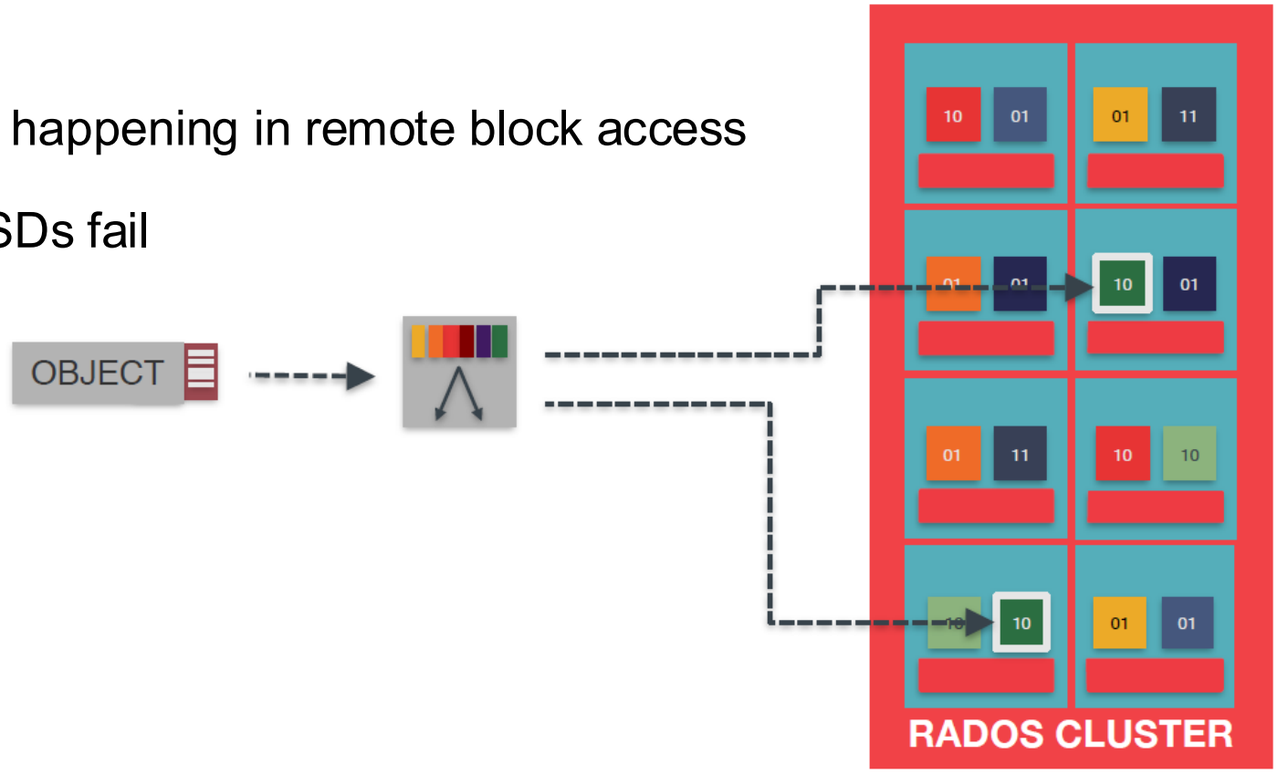
OSDs are aware of other OSDs and cluster status:

– overcoming the intense lookup over Ethernet as happening in remote block access

– helping with HA since data is replicated to let OSDs fail

Further features include:

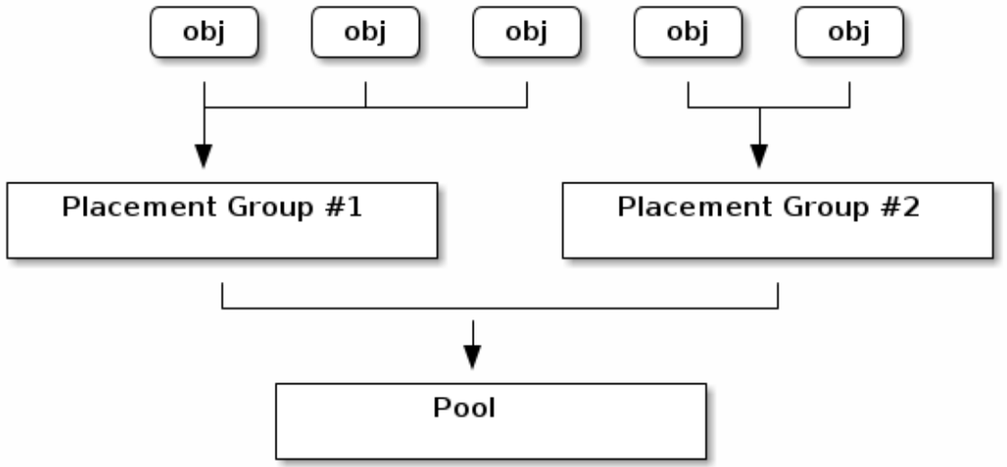
- Direct connect to OSD-clients
- Status of OSD is broadcasted
- Data is scrubbed / verified at all time
- Data is replicated to several sites

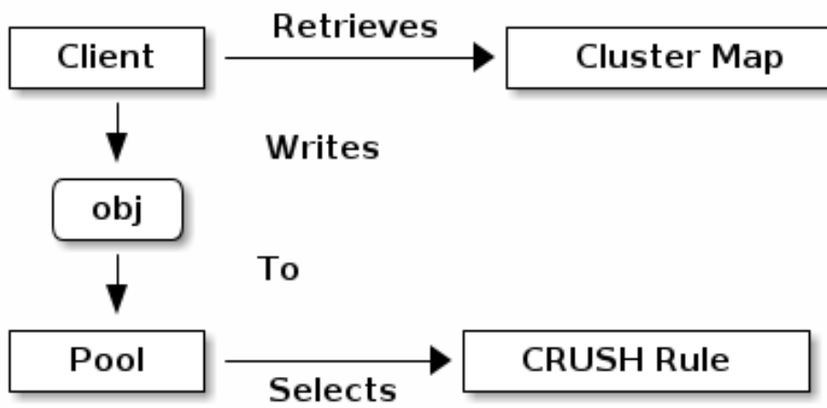


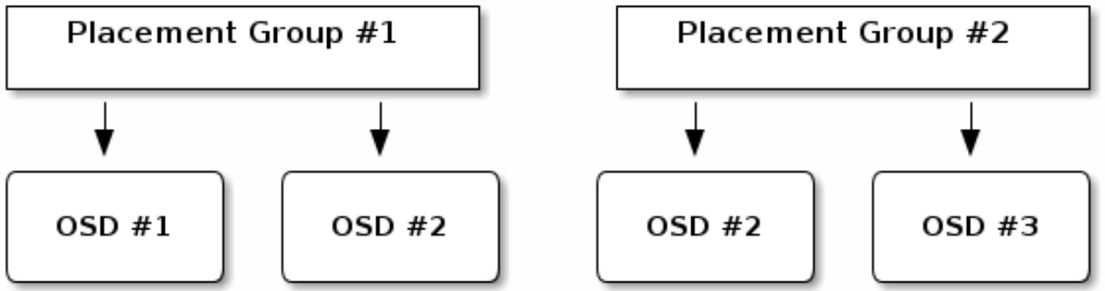
Mapping Data to OSDs

Placement Groups:

- Combine several OSDs
- Can be part of a larger tree/DAG-structure building up pools







CRUSH-Algorithm

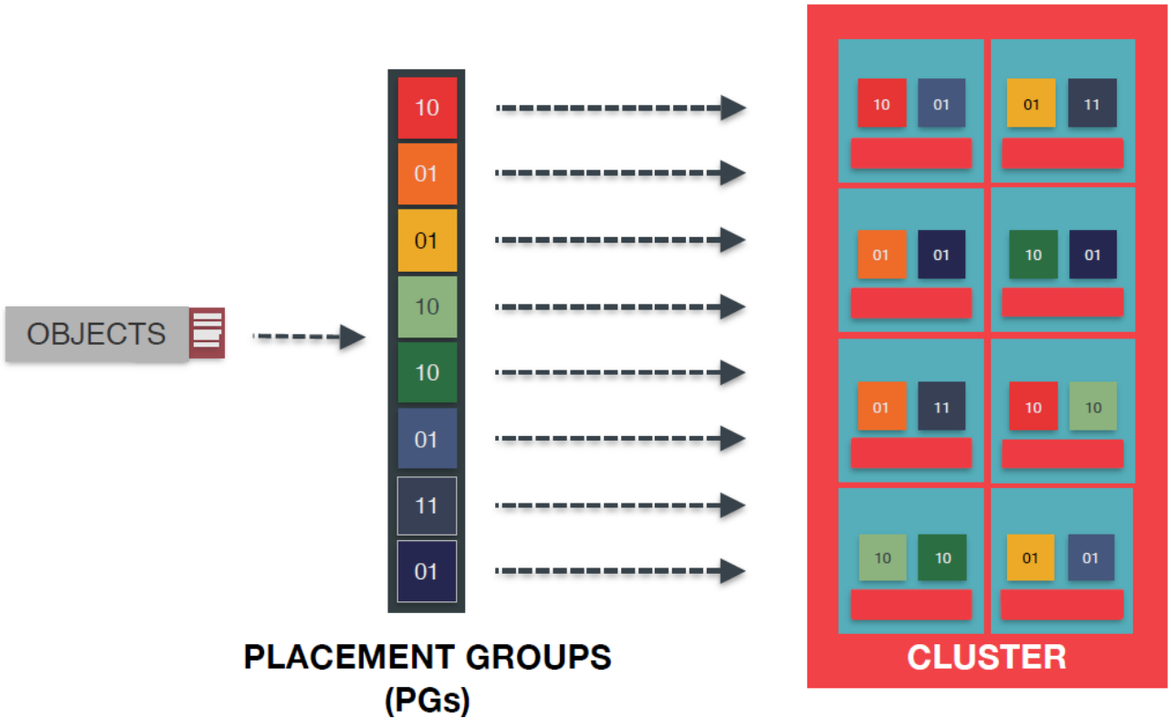
Based on:

- Clustermap, which is dezentralized / versioned
- Local computation of hashes / lookups

No centralized continuous lookup necessary!

Request to write object resulting in:

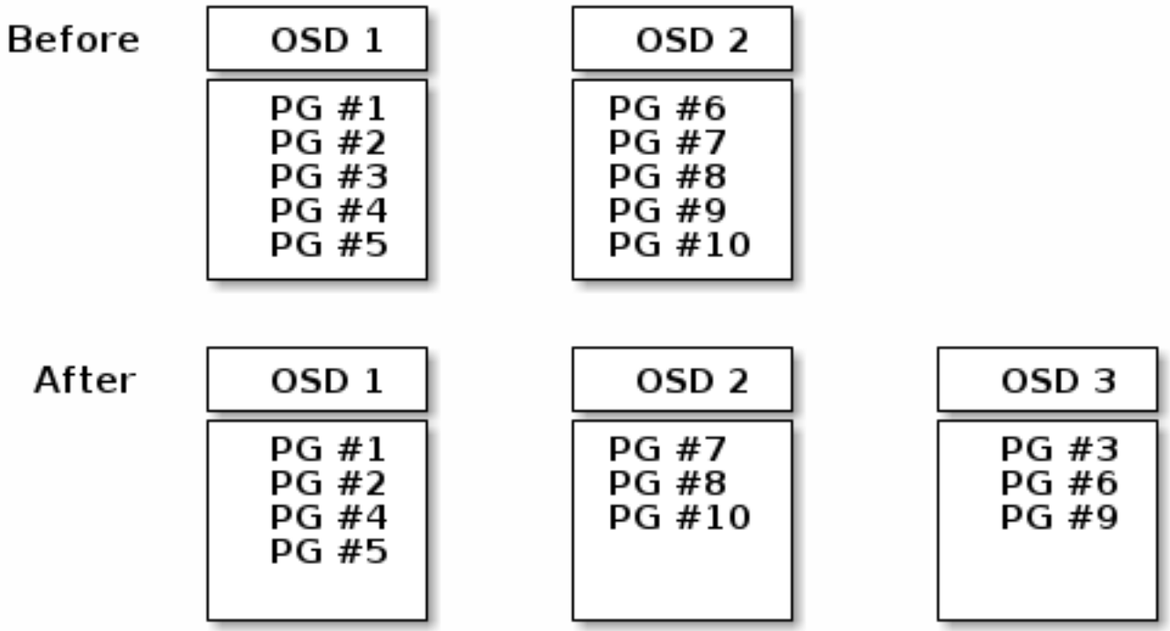
1. Hashing of object id
2. Lookup in cluster map to determine placement group
3. Lookup in placementgroup to determine OSDs



Modifying Disks

When OSDs are altered, rebalancing takes place:

- The Clustermap is altered and updated
- Based on Object ID, the new destination is computed according to the CRUSH-algorithm and the new Clustermap
- Data on Rest is transferred to the new OSD



CSI and K8s

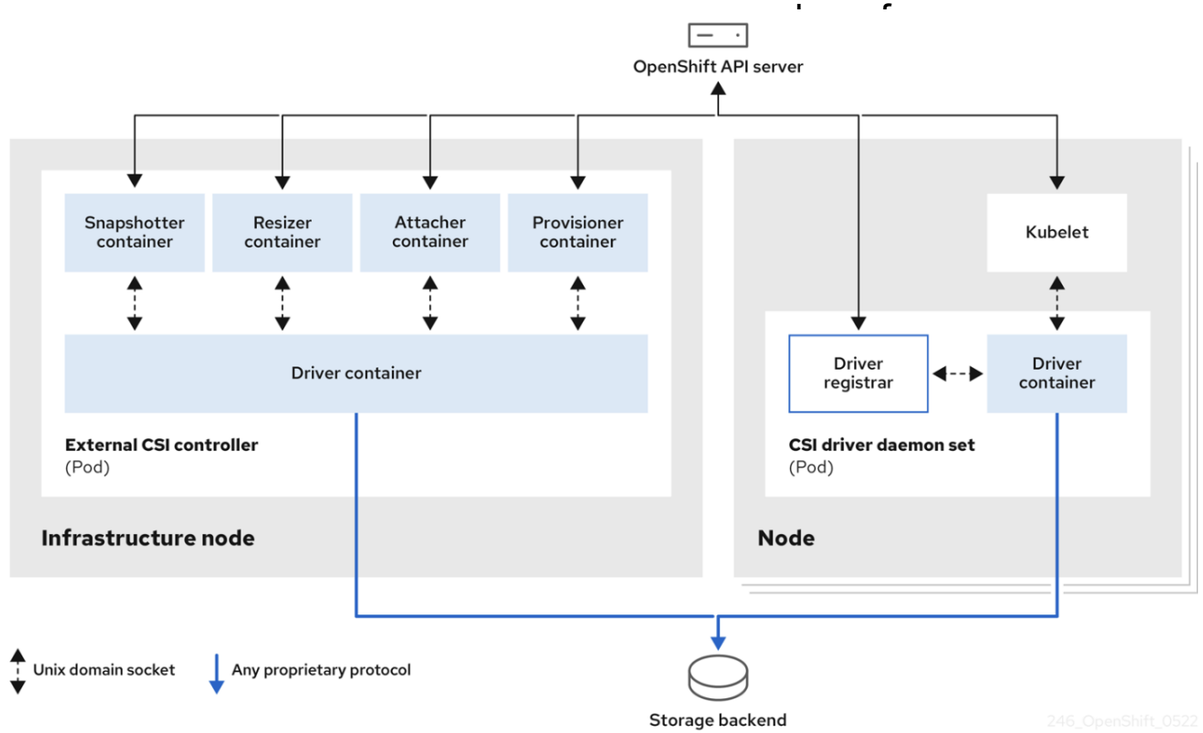
CSI abstracts the usage of proprietary storages in any Kubernetes Cluster offering:

- PVCs
- StorageClasses

Users of Kubernetes can get storage via PVC as:

- File Storage
- Block Storage

per default



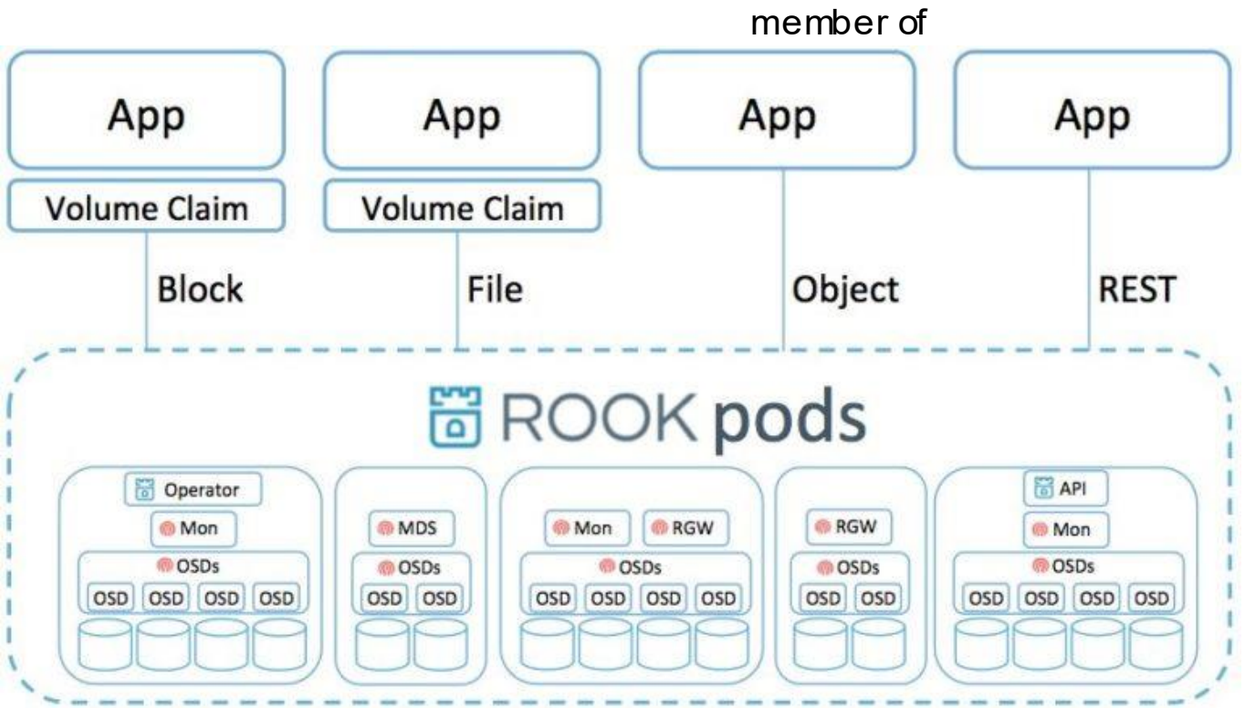
https://kubernetes-csi.github.io/docs/drivers.html

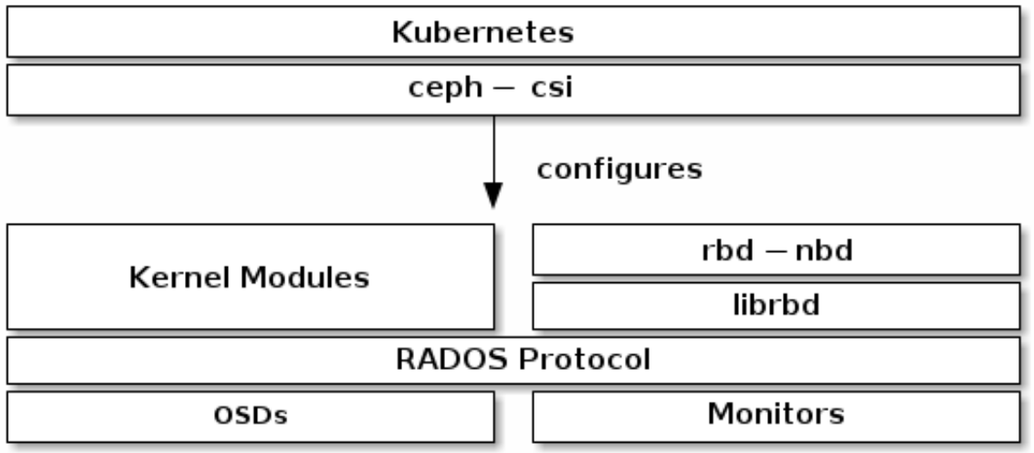
CephFS	cephfs.csi.ceph.com	v0.3, >=v1.0.0	A Container Storage Interface (CSI) Driver for CephFS	Persistent
Ceph RBD	rbd.csi.ceph.com	v0.3, >=v1.0.0	A Container Storage Interface (CSI) Driver for Ceph RBD	Persistent

Ceph and K8s → Rook

Ceph can also be deployed in Kubernetes abstracting / aligning the underlaying storage:

- All elements of Ceph are deployed as pods and maintained by an operator
- The “backend” is configured via `CephCluster`
- For the ”frontend”, all three interfaces are available: Block, File and Objects



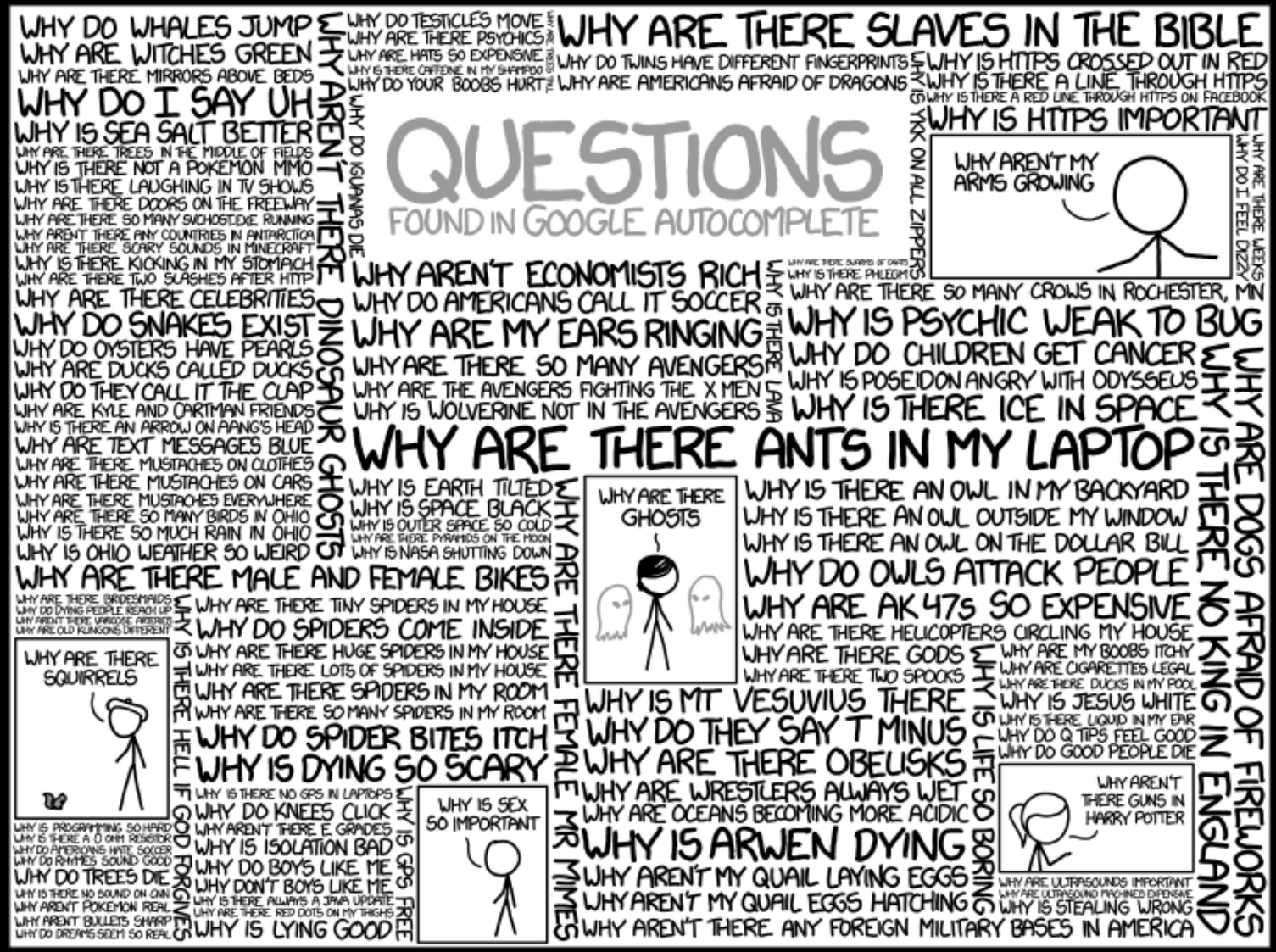


Summary

- Storage Platforms switch to virtualized platforms as well
  - Higher availability
  - More scalability
- Dedicated Storagelayers for dedicated use cases
  - Block Storage
  - File Storage
  - Object Storage

  However, borders are blurring

  (see Kubernetes StorageClass-Kinds)
- Platforms can be platforms for platform which can be platforms…


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SYCHOSTICKS RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REARM UP
WHY AREN'T THERE VARICOSE AIRBIKES
WHY ARE OLD KLINGONS DIFFERENT
WHY ARE THERE SQUIRRELS
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY IS YKK ON ALL ZIPPERS
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY DO IGUANAS DIE
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY ARE THERE SWARMS OF GNATS
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY IS THERE LAVA
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY IS SEX SO IMPORTANT
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY ARE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY IS LIFE SO BORING
WHY IS GPS FREE
WHY ARE THERE FEMALE MR MIMES
WHY IS THERE HELL IF GOD FORGIVES
WHY AREN'T THERE DINOSAUR GHOSTS

Summarizing Questions

- What are typical use cases of a block storage?
- What protocols work with object storages?
- What is the benefit of using software defined storage such as Ceph?
- What are the mechanisms that make Ceph resilient against failiing hardware?
- What is the purpose of an OSD in Ceph?